



DELHI PUBLIC SCHOOL NEWTOWN  
SESSION 2024-25  
HALF YEARLY EXAMINATION

CLASS: X  
SUBJECT: CHEMISTRY [SET A]

FULL MARKS: 80  
TIME: 2 HOURS

Maximum Marks: 80 Time allowed: Two hours

Answers to this Paper must be written on the paper provided separately. You will not be allowed to write during first 15 minutes.

This time is to be spent in reading the question paper.

The time given at the head of this Paper is the time allowed for writing the answers.

Section A is compulsory. Attempt any four questions from Section B. The intended marks for questions or parts of questions are given in brackets [].  
(This paper consists of seven printed pages)

SECTION A

*(Attempt all questions from this Section)*

Question 1

Choose the correct answers to the questions from the given options. [15]  
(Do not copy the questions, write the correct answers only.)

- (i) Rock salt is heated with concentrated sulphuric acid at a temperature above 200°C.

Which of the following could be the products:

P : Sodium sulphate.

Q: Hydrochloric acid.

R: Hydrogen chloride gas.

S : Sodium hydrogen sulphate.

(a) both P and R

(b) both R and S

(c) only P

(d) only R

- (ii) A compound M is heated in a test tube with sodium hydroxide solution. The gas evolved is bubbled through an aqueous solution of copper sulphate in excess and an inky blue solution is obtained.

Which of the following could be M?

(a) Ferrous chloride

(b) Ammonium chloride

(c) Zinc chloride

(d) Magnesium chloride

- (iii) Which of the following would occupy maximum volume at STP?

(a) 2 gram atoms of oxygen.

(b) 2 gram molecule of carbon dioxide

(c) 142 grams of chlorine

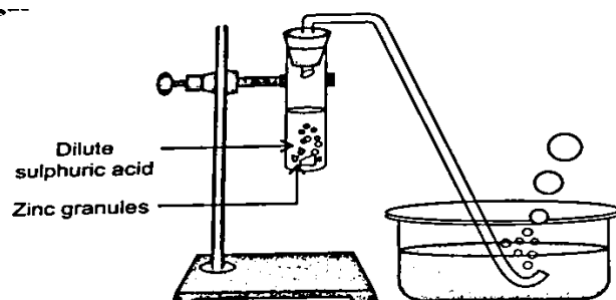
(d)  $6.023 \times 10^{23}$  molecules of ammonia

- (iv) Ammonia gas is passed over a black solid and a reddish brown residue is obtained. Which property of ammonia is exhibited in this reaction?
- Oxidizing agent
  - Basic gas
  - Supporter of combustion
  - Reducing agent
- (v) Assertion (A): Hydrogen chloride gas is not dried using calcium oxide.  
Reason (R): Hydrogen chloride is a basic gas.
- Both A and R are true, and R is the correct explanation of A.
  - Both A and R are true, but R is not the correct explanation of A.
  - A is true but R is false.
  - A is false but R is true.
- (vi) Which of the following has two lone pairs and two bond pairs?
- Water
  - Carbon dioxide
  - Ammonia
  - Nitrogen
- (vii) Element 'N' has electronic configuration 2, 8, 2. The number of oxygen atoms present in the oxide of 'N' is:
- 1
  - 2
  - 3
  - 4
- (viii)  $^{18}\text{O}$  is an isotope of oxygen. In the modern Periodic Table it will be placed:
- before oxygen
  - after oxygen
  - at the same position as oxygen
  - not have any position in the Periodic Table
- (ix) A nitrate which forms a precipitate with sodium hydroxide and is also soluble in excess of it:
- Ferrous nitrate
  - Ferric nitrate
  - Lead nitrate
  - Copper nitrate
- (x) Which of the following electronic configuration represents the most electronegative element?
- 2, 1
  - 2, 8, 2
  - 2, 8
  - 2, 7
- (xi) Assertion (A): Halogens do not form dinegative ions.  
Reason (R): After gaining one electron halogens achieve stable electronic configuration of noble gases.
- Both A and R are true, and R is the correct explanation of A.
  - Both A and R are true, but R is not the correct explanation of A.
  - A is true but R is false.
  - A is false but R is true.
- (xii) The ratio between the volumes occupied by 4.4 grams of nitrous oxide and 32 grams of oxygen at STP is: [O = 16, N = 14]
- 0.1:1
  - 0.2: 1
  - 1:1
  - 1:2
- (xiii) Copper oxide and manganese dioxide can be distinguished using:
- Ammonium hydroxide solution
  - Concentrated hydrochloric acid
  - Aqueous sodium hydroxide solution
  - Calcium hydroxide

(xiv) Which of the following about amphoteric oxides is correct?

- (a) Amphoteric oxides are water soluble
- (b) Amphoteric oxides are mostly covalent compounds
- (c) Amphoteric oxides can react with both acids and bases
- (d) Amphoteric oxides are mostly oxides of non-metals

(xv) Which property of sulphuric acid is demonstrated in the experiment set up shown below:

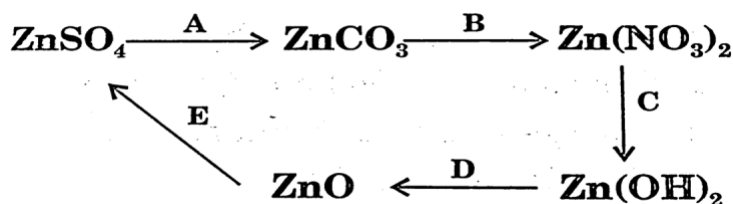


- (a) Non volatile acid
- (b) Oxidising agent
- (c) Dehydrating agent
- (d) Typical acid

### Question 2

(i) Write balanced chemical equations for the conversions A to E:

[5]



(ii) Match the following column A with column B:

[5]

| Column A                      | Column B                                               |
|-------------------------------|--------------------------------------------------------|
| (a) Lead                      | 1. consists of molecules                               |
| (b) Sulphuric acid            | 2. zero electron affinity                              |
| (c) Neon                      | 3. produces hydrogen on reacting with Sodium hydroxide |
| (d) Calcium chloride solution | 4. dibasic acid                                        |
| (e) Carbon tetrachloride      | 5. no precipitate with liquor ammonia                  |

(iii) Complete the following by choosing the correct answers from the bracket:

[5]

- (a) If an element has seven electrons in the valence shell, then it is likely to have the \_\_\_\_\_ [smallest/ largest] atomic radius amongst all the elements in that same period.
- (b) \_\_\_\_\_ [sulphuric acid/ nitric acid] does not form an acid salt.
- (c) A \_\_\_\_\_ [reddish brown/ dirty green] coloured precipitate is formed when sodium hydroxide is added to a solution of ferric chloride.
- (d) Ionic compounds undergo \_\_\_\_\_ [ionization/ dissociation] in molten state.
- (e) An \_\_\_\_\_ [acidic/ alkaline] solution will turn phenolphthalein solution pink.

(iv) Write the suitable term for the following:

[5]

- (a) A pair of valence electrons which does not take part in covalent bond formation.
- (b) A salt formed by incomplete neutralization of a base by an acid.

- (c) The volume occupied by one mole of a gas at STP.
  - (d) The energy released when a neutral gaseous atom gains electron in its valence shell.
  - (e) A solution which boils at a fixed temperature without change in concentration.
- (v) (a) Draw the electron dot and cross diagram for the following: [5]
1. Methane
  2. Ammonium ion
  3. Magnesium oxide
- (b) Define the following:
1. pH
  2. Electronegativity

**SECTION B (40 Marks)**  
(Attempt any four questions)

**Question 3**

- (i) Identify the reactant and write the balanced equation for the following: [2]  
Nitric acid reacts with compound M to give a blue solution, water and carbon dioxide.
- (ii) What property of Sulphuric acid is exhibited in each of the following cases: [2]
- (a) In the preparation of nitric acid when it reacts with Sodium nitrate.
  - (b) Concentrated sulphuric acid reacts with Zinc to produce Sulphur dioxide.
- (iii) The ionization potential of an element X is greater than that of element Y. [3]
- (a) How is the reducing power of X likely to compare with that of Y?
  - (b) How is the atomic radius of X likely to compare with that of Y?
  - (c) State whether X is likely to be placed to the right or to the left of Y in the periodic table.
- (iv) You are provided with the list of chemicals mentioned below in the box: [3]

Nitric acid, potassium hydroxide solution, copper carbonate, zinc, chlorine, hydrochloric acid, copper, dilute hydrochloric acid, iron, sulphuric acid

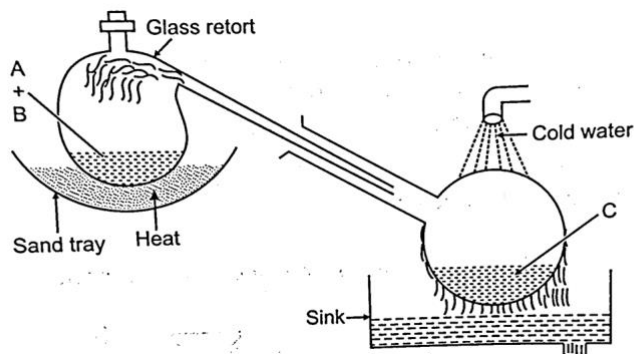
Using suitable chemicals from the list given, write balanced chemical equation for the preparation of salts mentioned below: [3]

- (a) Copper nitrate
- (b) Potassium zincate
- (c) Ferrous sulphate

**Question 4**

- (i) The following questions relate to the industrial preparation of ammonia: [2]
- (a) Write equation for the catalyzed reaction.
  - (b) How is ammonia gas separated from the unused reactants?
- (ii) On analyzing an organic sample weighing 10.47 grams, Rahul found the sample contains 6.21 grams of metal X (atomic weight 207) and 4.26 gram of non-metal Y (atomic weight 35.5). Determine the empirical formula of the compound. [2]
- (iii) Give balanced equation for each of the following: [3]
- (a) Action of warm water on Calcium nitride.
  - (b) Oxidation of sulphur with nitric acid.
  - (c) Excess of ammonia is reacted with chlorine.

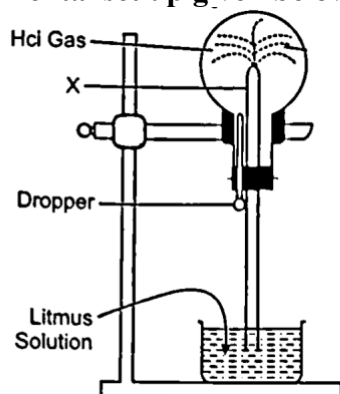
- (iv) The diagram below is a representation of the laboratory preparation of Nitric acid. With respect to the process answer the following questions: [3]



- Identify solid B
- An all-glass apparatus is used, why?
- Why is the temperature kept below  $200^{\circ}\text{C}$ ?

#### Question 5

- Aparna passed a mixture of dry ammonia gas and oxygen gas over heated platinum at a temperature of  $700^{\circ}\text{C}$ . What did she observe?
  - Write a balanced chemical equation for the above reaction. [2]
- Arrange the following as instructed: [2]
  - Methane, water, ammonia (decreasing order of number of single bonds)
  - Nitrogen, boron, fluorine (increasing order of number of valence electrons)
- Rohit was given a salt 'X' which was white in colour for analysis. On strong heating it produced a residue which is black in colour, a colourless gas and a reddish brown gas. The solution of salt 'X' when tested with excess of ammonium hydroxide produced an inky blue solution.
  - Name the coloured gas evolved when Rohit heated the salt strongly.
  - Which cation was present in the salt given to Rohit?
  - Identify the salt given to Rohit for analysis. [3]
- Study the experimental set up given below and answer the following questions: [3]

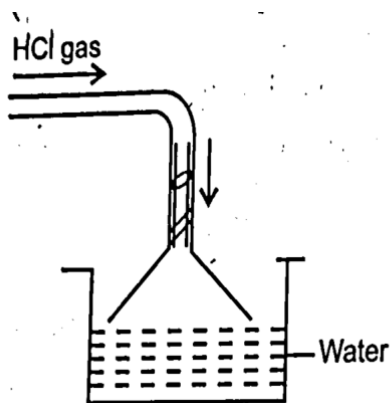


- Name the experiment.
- Which property of hydrogen chloride gas is demonstrated by this experiment?
- Name another gas which has the same property and can be demonstrated through this experiment.

#### Question 6

- Element 'Y' forms a chloride with formula  $\text{XCl}_3$ , which is a solid with high melting point. 'Y' would most likely be placed in the group of the Periodic Table as:
  - (a) Li (b) Ca (c) B (d) C
  - Justify your answer in the above question (1). [2]

- (ii) A student prepared hydrochloric acid in the laboratory by setting up the experiment given below: [2]



- (a) Name the arrangement.  
 (b) Give 2 reasons why the above set up is needed.  
 (iii) The gases chlorine, nitrogen, ammonia and sulphur dioxide are collected under the same conditions of temperature and pressure. The following table gives the volumes of gases collected and the number of molecules (X) in 20 litres of nitrogen. Copy and complete the following table giving the number of molecules in the other gases in terms of X. [3]

| Gas             | Volume (litres) | Number of molecules |
|-----------------|-----------------|---------------------|
| Chlorine        | 10              |                     |
| Nitrogen        | 20              | X                   |
| Ammonia         | 20              |                     |
| Sulphur dioxide | 5               |                     |

- (iv) X [ 2,7] and Y [ 2,8,1]. Using this information complete the following: [3]  
 (a) \_\_\_\_\_ is the non-metallic element.  
 (b) Metal atoms tend to have maximum of \_\_\_\_\_ valence electrons.  
 (c) \_\_\_\_\_ is the reducing agent.

#### Question 7

- (i) 300 cc of Ethane burns in 1250 cc of oxygen to form carbon dioxide and water vapour according to equation.  

$$2\text{C}_2\text{H}_6(\text{g}) + 7\text{O}_2(\text{g}) \rightarrow 4\text{CO}_2(\text{g}) + 6\text{H}_2\text{O}(\text{l})$$
 Calculate:  
 (a) The volume of carbon dioxide formed.  
 (b) The volume of unused oxygen.  
 (c) State the law used for the above calculations. [3]
- (ii) Raman has solutions X, Y and Z that has pH 12, 7 and 1 respectively. Which solution: [3]  
 (a) will liberate a gas with rotten egg smell, when added to ferrous sulphide.  
 (b) will turn pH paper green.  
 (c) will produce a pungent smelling basic gas when heated with ammonium sulphate.
- (iii) Concentrated nitric acid oxidizes phosphorus according to the following equation: [4]  

$$\text{P} + 5\text{HNO}_3 \rightarrow \text{H}_3\text{PO}_4 + 5\text{NO}_2 + \text{H}_2\text{O}$$
 (a) What mass of phosphoric acid can be prepared from 3.1 g of phosphorus?  
 (b) What mass of nitric acid will react with 3.1 g of phosphorus?  
 (c) What volume of nitrogen dioxide formed when 3.1 g of phosphorus reacts with nitric acid?  
 (d) Calculate the number of moles in 3.1 g of phosphorus.

**Question 8**

- (i) Give reason for the following: [2]
- (a) Ammonium nitrate is not used for the laboratory preparation of ammonia.
  - (b) Sodium hydroxide solution cannot be used to distinguish between lead nitrate and zinc nitrate solution.
- (ii) Compare the properties of sodium chloride and carbon tetrachloride based on: [2]
- (a) Solubility in water.
  - (b) Electrical conductivity
- (iii) Identify the salts P, Q, R from the following observations: [3]
- (a) Salt P has is a green solid. On heating leaves a black residue. Salt P produces a colourless, odourless gas with brisk effervescence on addition of dilute sulphuric acid, the produced gas turns lime water milky and has no effect on acidified potassium dichromate solution.
  - (b) Salt Q is a white solid which gives a non-persistent lilac coloured flame. A curdy white precipitate is formed on adding silver nitrate solution to an aqueous solution of Q.
  - (c) Salt R is a white crystalline solid which imparts a golden yellow colour to the flame. On adding barium chloride solution to an aqueous solution of R a dense white precipitate is formed insoluble in mineral acids.
- (iv) With respect to brown ring test for nitrates, answer the following: [3]
- (a) Write the name and formula of the brown ring formed.
  - (b) Explain why freshly prepared ferrous sulphate solution is used for this test.